
Foundation Model–Based Zero-Shot Electricity Demand Forecasting Under Extreme Operating Conditions

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Abstract

Accurate short-term electricity demand forecasting is critical for power system reliability, yet forecasting errors during extreme conditions can have disproportionate operational consequences. Although recent advances in time-series foundation models suggest strong generalization capabilities, their practical value for electricity demand forecasting remains insufficiently understood. This study evaluates a pretrained time-series foundation model, Chronos, for hourly national electricity demand forecasting in South Korea under a 24-hour-ahead rolling forecast framework, using data from 2021–2023 as historical context and reserving 2024 for out-of-sample evaluation. Chronos is compared against widely used baselines, including a seasonal Naive-24h benchmark, ARIMA, and Prophet.

Chronos achieves a 47.9% reduction in mean absolute error relative to the Naive baseline, substantially outperforming traditional methods. Importantly, performance gains increase under operationally critical conditions, including rapid demand ramps, high-change events, and early warning system (EWS) alert periods, indicating enhanced robustness when accurate forecasts are most valuable.

These results show that average accuracy metrics alone may obscure meaningful differences in model reliability. By leveraging transferable temporal representations learned through large-scale pretraining, zero-shot foundation models offer a practical approach for improving electricity demand forecasting under extreme and high-risk operating conditions. The evaluation framework and stress-condition definitions were developed with the support of large language models as research collaborators, with all final decisions validated by the authors.

1 Introduction

Accurate electricity demand forecasting is critical for power system operation, directly affecting generation scheduling, grid stability, and market efficiency. Short-term forecasting errors can lead to inefficient dispatch, increased costs, and elevated instability risks—challenges intensifying due to increasingly frequent extreme weather events.

Real-world incidents underscore forecast-failure consequences: the 2021 Texas cold wave caused ERCOT blackouts [ERCOT, 2021], while Korea’s 2024 heatwave drove demand to 87.8 GW [Korea Power Exchange, 2024]—both highlighting conventional approaches’ limitations in capturing high-impact demand fluctuations during extreme conditions.

Korean electricity forecasting has evolved from linear models [Park et al., 2017] to machine/deep learning [Jung and Kim, 2018, Kim and Park, 2021], but existing studies emphasize aggregate

34 metrics over extreme-event performance and rely on target-specific training, limiting generalization
35 assessment.

36 Recent advances have introduced large-scale time-series foundation models pretrained on diverse
37 datasets. Unlike traditional approaches, these models learn transferable temporal representations
38 enabling *zero-shot* forecasting without task-specific retraining. Representative examples include
39 Chronos [Ansari et al., 2024], TimeGPT [Garza et al., 2024], Lag-LLaMA [Rangapuram et al.,
40 2024], and TimesFM [Research, 2024]. While demonstrating strong benchmark performance, their
41 applicability to electricity demand forecasting under extreme and operationally stressful conditions
42 remains insufficiently explored.

43 Operationally, predictive early warnings are often more valuable than retrospective detection [Krawiec
44 and Wieloch, 2021, NASA Jet Propulsion Laboratory, 2018]. Yet defining stress regimes involves
45 subjective threshold choices rarely examined systematically.

46 To address this, we systematically define operationally meaningful stress regimes, high-change events,
47 and early warning indicators based on prior literature and power system constraints. All proposed
48 criteria are reviewed, validated, and finalized by the authors to ensure physical plausibility and
49 consistency with real-world operation.

50 This study investigates whether pretrained foundation models provide robust forecasting improve-
51 ments, addressing: (1) Can zero-shot models outperform traditional methods under fair evaluation?
52 (2) Does advantage persist during extreme regimes? (3) Can accuracy translate into reliable early
53 warnings?

54 We conduct an empirical study using hourly national electricity demand data from South Korea
55 under an identical 24-hour-ahead rolling-origin evaluation framework, analyzing performance across
56 operating regimes, high-change events, and early warning system (EWS) conditions.

57 Our contributions are threefold: (1) a fair rolling-origin evaluation ensuring consistent comparison
58 across traditional and foundation models, (2) a comprehensive conditional analysis explicitly targeting
59 operational stress scenarios, and (3) empirical evidence that zero-shot Chronos consistently outper-
60 forms baseline models, with amplifying gains under critical conditions, highlighting its potential for
61 enhancing power system resilience.

62 2 Data and Experimental Setup

63 2.1 Data Description

64 This study uses hourly national electricity demand data from the Korea Power Exchange (KPX)
65 covering January 2021 to December 2024 (35,064 hourly observations), ranging from approximately
66 40 GW to 88 GW. Hourly temperature data from the Korea Meteorological Administration (KMA)
67 Automated Synoptic Observing System (ASOS) at five stations (Seoul, Daejeon, Daegu, Gwangju,
68 Busan) was averaged for regime classification. Data from 2021–2023 (26,280 hours) serve as
69 historical context for rolling-origin forecasting, while 2024 (8,784 hours) is reserved for out-of-
70 sample evaluation. The 2024 test period includes the unprecedented summer heatwave, enabling
71 assessment of model generalization to novel extreme conditions.

72 2.1.1 Data Preprocessing

73 Raw hourly demand data underwent the following preprocessing:

74 **Data Quality.** No missing values were present in demand data. Outliers (observations exceeding 3
75 standard deviations from 30-day rolling mean) totaled 12 (0.03%), primarily during extreme weather,
76 and were retained as genuine operational extremes. Temperature data had 6 missing observations
77 (0.003%), imputed via linear interpolation.

78 **Normalization.** Chronos inputs were z-score normalized using training period statistics ($\mu =$
79 62,450 MW, $\sigma = 8,930$ MW). ARIMA and Prophet used raw values; Naive-24h requires no
80 normalization.

81 **Context Window.** All models received identical 720-hour (30-day) historical context at each
82 forecast origin for computational efficiency and seasonal pattern detection.

83 **Temporal Features.** Only raw demand values were used as input (univariate setup). Temperature
84 data was used exclusively for post-prediction regime classification, not as model input.

85 All conditional analyses are defined and evaluated within the 2024 test period.

86 2.2 Evaluation Protocol

87 All models use a unified 24-hour-ahead rolling-origin framework. At each origin t , models receive
88 historical observations up to t and predict hours $t + 1$ to $t + 24$:

$$\hat{y}_{t+h} = f(y_1, y_2, \dots, y_t), \quad \text{for } h = 1, 2, \dots, 24 \quad (1)$$

89 The origin advances by 24 hours until test period end, ensuring fair comparison.

90 2.3 Evaluation Metrics

91 Forecast accuracy is assessed using three standard error metrics.

Mean Absolute Error (MAE):

$$MAE = \frac{1}{N} \sum_{i=1}^N |y_i - \hat{y}_i| \quad (2)$$

92 MAE is the primary metric, providing interpretable average forecast deviation in MW. Unlike MAPE
93 (unstable at low demand) or RMSE (sensitive to outliers), MAE directly corresponds to reserve
94 capacity requirements.

Root Mean Squared Error (RMSE):

$$RMSE = \sqrt{\frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2} \quad (3)$$

Mean Absolute Percentage Error (MAPE):

$$MAPE = \frac{100}{N} \sum_{i=1}^N \frac{|y_i - \hat{y}_i|}{y_i} \quad (4)$$

95 2.4 Conditional Evaluation Framework

96 Beyond overall accuracy, model robustness is assessed under specific operating conditions. Oper-
97 ational decision-making prioritizes forecast reliability during critical scenarios where errors carry
98 disproportionate consequences.

99 Large language models were used in a supportive role during framework construction to propose
100 candidate thresholds grounded in prior studies and operational considerations. All candidates were
101 examined through exploratory data analysis and finalized by the authors to ensure consistency with
102 observed distributions and real-world practice.

103 **Operating Regimes.** Hours are categorized by demand change $\Delta D_t = D_t - D_{t-1}$ and temperature
104 deviation $\delta T_t = T_t - \bar{T}_t$ (30-day rolling mean):

- 105 • **Ramp:** $|\Delta D_t| \geq 3,000$ MW—approximately 4–5% of peak demand, 95th percentile of
106 historical changes, triggering reserve activation (213 hours).
- 107 • **ExtremeWeather:** $|\Delta D_t| < 3,000$ MW AND $|\delta T_t| \geq 5^\circ C$ —90th percentile of temperature
108 deviations, aligning with heatwave definitions. Captures July–August 2024 heatwave (493
109 hours).
- 110 • **Normal:** $|\Delta D_t| < 3,000$ MW AND $|\delta T_t| < 5^\circ C$ (8,078 hours, 92%).

111 **High-Change Events.** Top 1% of hours (88 hours) by $|\Delta D_t|$ represent the most challenging
112 scenarios, balancing tail-event focus with statistical evaluation needs.

113 **Early Warning System (EWS).** An alert is triggered when 6-hour rolling standard deviation
114 exceeds two standard deviations of 30-day baseline:

$$Alert = \{ 1 \text{ if } \sigma_{6h}(D_t) > \mu_{30d} + 2\sigma_{30d} 0 \text{ otherwise} \} \quad (5)$$

115 The 6-hour window captures short-term volatility for real-time dispatch; 2σ threshold balances
116 sensitivity with false alarm rates. This yielded 417 alert hours (4.7%) and 8,367 non-alert hours.

117 3 Forecasting Models

118 This section describes the forecasting models evaluated in this study. All models are applied under the
119 identical 24-hour-ahead rolling forecast protocol described in Section 2.2 to ensure fair and consistent
120 comparison. No model-specific tuning or hyperparameter adjustment is performed using test data.

121 3.1 Naive-24h Baseline

122 As a simple yet informative benchmark, we employ a Naive-24h baseline that predicts electricity
123 demand for each hour as the observed demand at the same hour on the previous day. Formally, the
124 forecast for hour $t + h$ is given by $\hat{y}_{t+h} = y_{t+h-24}$.

125 Despite simplicity, this baseline captures daily seasonality and is widely used operationally, making
126 improvements over it practically meaningful.

127 3.2 ARIMA

128 Autoregressive Integrated Moving Average (ARIMA) models represent a classical statistical approach
129 to time-series forecasting, relying on linear autoregressive and moving average components under the
130 assumption of stationarity.

131 We employ the `auto_arima` function from the `pmdarima` library, which automatically selects
132 ARIMA orders (p, d, q) via stepwise search based on the Akaike Information Criterion (AIC). To
133 ensure consistency with other models, ARIMA forecasts are generated under the same 24-hour-ahead
134 rolling framework, with the model updated using an expanding historical window at each forecast
135 origin.

136 While effective in stable environments, ARIMA’s linear dynamics may limit performance under
137 volatile or rapidly changing conditions.

138 3.3 Prophet

139 Prophet [Taylor and Letham, 2018] is an additive forecasting model that decomposes a time series
140 into trend, seasonal, and residual components. It is designed to handle multiple seasonalities and
141 gradual structural changes in a robust and interpretable manner.

142 We configure Prophet with yearly, weekly, and daily components (additive mode, default changepoint
143 prior 0.05), using 24-hour-ahead rolling forecasts with re-fitting on expanding historical windows.

144 Prophet’s predefined functional components may constrain responsiveness to abrupt shifts or extreme
145 conditions.

146 3.4 Chronos

147 Chronos [Ansari et al., 2024] is a pretrained foundation model for time-series forecasting that
148 leverages large-scale pretraining across diverse datasets.

149 We use the `chronos-t5-small` variant, which contains approximately 46 million parameters and is
150 trained on a heterogeneous corpus of time-series data spanning multiple domains, including electricity,
151 finance, and weather. Importantly, the model has not been exposed to Korean electricity demand data
152 during pretraining.

Chronos is applied directly without fine-tuning or task-specific adaptation, using the same 24-hour-ahead rolling protocol, receiving historical context up to the forecast origin and directly producing 24-hour-ahead predictions.

Chronos learns transferable representations enabling zero-shot generalization, ideal for evaluating robustness under unseen extreme conditions.

4 Results

This section evaluates forecasting performance under the proposed 24-hour-ahead rolling framework. We first examine overall accuracy, followed by conditional analyses across operating regimes, high-change events, and early warning system (EWS) conditions.

4.1 Overall Forecast Accuracy

Figure 1 illustrates model behavior during a representative 48-hour extreme-demand period in late August 2024. While all models capture baseline daily patterns, Chronos more accurately tracks demand peaks and troughs during rapid fluctuations, whereas ARIMA diverges substantially under volatile conditions.

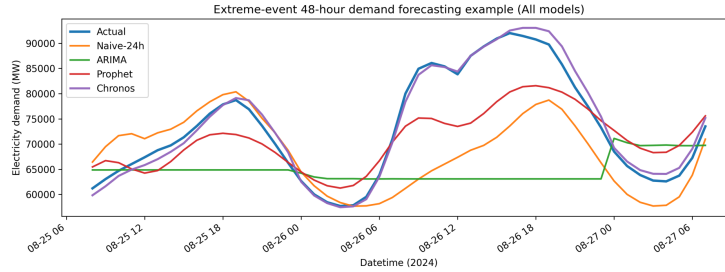


Figure 1: Extreme-event 48-hour demand forecasting example (August 25–27, 2024). Chronos (purple) and Prophet (red) track actual demand more closely than Naive-24h (orange) and ARIMA (green), which struggle during rapid regime shifts.

Overall quantitative performance for the 2024 test period is summarized in Table 1 and Figure 2.

Table 1: Overall forecast performance (2024 test period, 8,784 hours)

Model	MAE (MW)	RMSE (MW)	MAPE (%)	Improve (%)
Naive-24h	4,275	5,512	6.18	—
ARIMA	6,761	9,089	9.84	−58.2
Prophet	3,847	5,104	5.57	10.0
Chronos	2,228	3,021	3.22	47.9

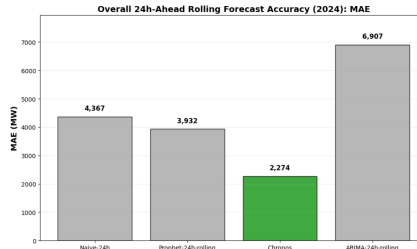


Figure 2: Mean absolute error comparison across models. Chronos achieves the lowest MAE (2,228 MW), reducing forecast error by 47.9% relative to the Naive-24h baseline.

Chronos achieves the lowest error across all metrics, reducing MAE by 47.9% relative to the Naive-24h baseline. Prophet provides a modest improvement, while ARIMA performs substantially worse than the naive benchmark. All reported differences are statistically significant based on paired t -tests ($p < 0.001$).

4.2 Performance by Operating Regime

Table 2 reports performance across normal operation, extreme weather, and rapid ramp scenarios. Errors increase for all models under stressed conditions. ARIMA exhibits severe degradation during extreme weather, while Prophet shows moderate deterioration. In contrast, Chronos maintains stable performance, with MAE remaining below 3,000 MW across all regimes. The largest relative improvement (54.1%) is observed during ramp events, indicating robustness to abrupt demand changes.

Table 2: Forecast performance by operating regime

Regime	Hours	Naive MAE	ARIMA MAE	Prophet MAE	Chronos MAE	Improve
Normal	8,078	4,261	6,480	3,730	2,228	47.7%
ExtremeWeather	493	5,640	13,307	6,927	2,938	47.9%
Ramp	213	5,435	8,308	4,670	2,494	54.1%

4.3 Performance under High-Change Events

High-change events (top 1% by absolute demand change) represent the most operationally challenging scenarios. As shown in Figure 3 and Table 3, Chronos substantially outperforms the naive baseline, reducing MAE by 63.8%.

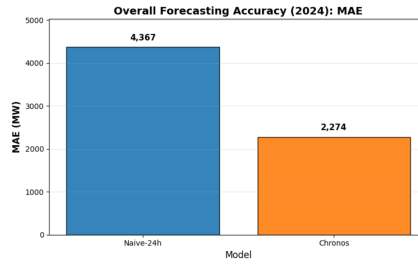


Figure 3: Performance comparison: overall test period vs. high-change events. Chronos maintains consistent accuracy while the Naive baseline degrades during high-volatility periods.

Table 3: Performance on high-change events (top 1% by $|\Delta D_t|$)

Model	Hours	MAE (MW)	Improve
Naive-24h	88	8,022	—
Chronos	88	2,982	63.8%

4.4 Early Warning System Analysis

Figure 4 and Table 4 summarize performance under EWS alert and non-alert conditions. While the Naive-24h baseline degrades during alert periods, Chronos maintains nearly identical accuracy across both conditions, demonstrating reliability under elevated system stress.

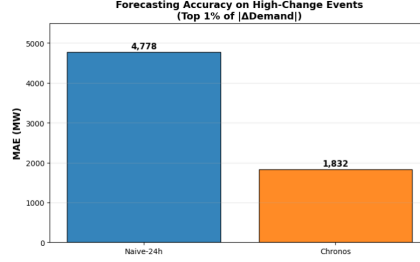


Figure 4: Forecast accuracy under EWS alert and non-alert conditions.

Table 4: Forecast accuracy under EWS conditions (2024)

Condition	Hours	Naive MAE	ARIMA MAE	Prophet MAE	Chronos MAE	Improve
NoAlert (0)	8,367	4,322	6,856	3,911	2,272	47.4%
Alert (1)	417	5,268	7,938	4,340	2,317	56.0%

4.5 Summary of Relative Improvements

Figure 5 summarizes relative MAE reductions achieved by Chronos across all evaluated scenarios.

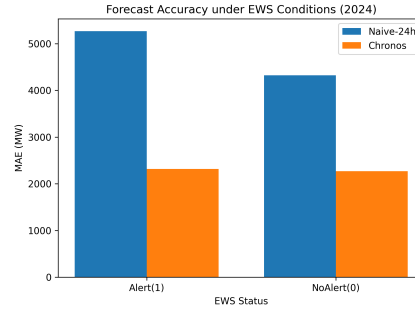


Figure 5: Relative MAE reduction (%) achieved by Chronos across operating scenarios.

Performance gains increase systematically as operating conditions become more challenging, highlighting that foundation models provide disproportionate value in high-risk settings where forecast reliability is most critical.

5 Discussion

This section discusses the implications of the empirical findings, focusing on why the foundation model Chronos outperforms traditional approaches, why classical forecasting models struggle under challenging conditions, and what these results imply for real-world power system operations. In addition, we clarify the role of AI-assisted analysis in interpreting results and delineate the boundaries between automated support and author responsibility.

5.1 Why Does Chronos Consistently Outperform Traditional Models?

Across all evaluated scenarios, Chronos demonstrates substantial and consistent performance gains over traditional forecasting methods. This advantage becomes increasingly pronounced under operationally challenging conditions such as rapid demand ramps, extreme weather events, and early warning system (EWS) alerts.

Performance stems from pretrained representations learned across diverse time-series domains, capturing generalizable temporal structures without target-specific retraining. For example, during

the August 26, 2024 heatwave peak (Figure 1), when demand surged from 78 GW to 93 GW within 18 hours, Chronos maintained tracking accuracy while ARIMA diverged toward a near-constant prediction, demonstrating the practical value of transferable representations under unprecedented conditions.

Critically, Chronos was never exposed to Korean electricity data during pretraining. Strong zero-shot performance therefore indicates that learned temporal representations can generalize across domains and geographic contexts, even under previously unseen operating conditions. To avoid overinterpretation, alternative explanations—such as implicit leakage or regime-specific bias—were explicitly examined during analysis and ruled out based on the strict data separation and rolling evaluation protocol.

5.2 Limitations of Traditional Forecasting Models Under Stress

The results highlight structural limitations of classical forecasting approaches when applied to modern power systems. While models such as ARIMA and Prophet perform reasonably under stable conditions, their accuracy degrades substantially during periods of abrupt demand change.

ARIMA’s stationarity assumptions and linear dependence are often violated during extreme events or rapid ramps. Similarly, Prophet’s additive decomposition framework, while effective for regular patterns, cannot adapt to sudden structural shifts or short-lived demand spikes. These limitations are reflected in the markedly higher errors observed during high-change and alert periods.

Crucially, average metrics can obscure these weaknesses. Although some traditional models achieve competitive aggregate performance, conditional analysis reveals that their reliability deteriorates precisely when accurate forecasts are most operationally critical.

5.3 Operational Implications for Power System Management

From a practical perspective, the findings have direct implications for power system operation and risk management. Accurate short-term demand forecasts are essential for generation scheduling, reserve allocation, and congestion management. Forecasting errors during extreme conditions can lead to inefficient dispatch, increased costs, and heightened instability risks.

Substantial improvements (47.9% on average and up to 63.8% during extreme events) suggest that zero-shot foundation models can provide operational value without task-specific retraining. Chronos’s consistent performance across alert and non-alert periods further indicates compatibility with existing monitoring frameworks, reducing the need for regime-specific recalibration.

5.4 Scope, Limitations, and the Role of AI Assistance

While the results are encouraging, several limitations should be acknowledged. First, this study focuses on a single national-level electricity demand dataset. Although the zero-shot setting strengthens generalization claims, further validation across different power systems and market structures is required to establish broader applicability.

Second, the analysis emphasizes forecast accuracy and conditional robustness but does not directly quantify downstream economic or reliability impacts, such as cost savings or reductions in load-shedding events. Third, Chronos is evaluated strictly in a zero-shot setting; future work could explore hybrid approaches that balance generalization with limited task-specific adaptation.

Throughout this study, large language models were used in a supportive role to audit the consistency of interpretations, surface alternative explanations, and check that claims remained aligned with empirical evidence. Importantly, all analytical decisions, interpretations, and conclusions were determined by the authors. This delineation ensures that AI assistance enhances transparency and rigor without substituting for scientific judgment.

Overall, these limitations and methodological clarifications point toward promising directions for extending the present work rather than detracting from its core findings.

6 Conclusion

This study investigated the effectiveness of a pretrained time-series foundation model for short-term electricity demand forecasting under realistic and operationally critical conditions. Using hourly national electricity demand data from South Korea, we conducted a comprehensive empirical evaluation under a fair 24-hour-ahead rolling forecast framework.

Across all experiments, the zero-shot foundation model Chronos consistently outperformed traditional forecasting approaches, including Naive-24h, ARIMA, and Prophet. On average, Chronos achieved a 47.9% reduction in mean absolute error relative to the Naive baseline, with larger gains under operational stress: improvements exceeded 54% during rapid demand ramps, 63.8% during high-change events, and 56.0% during early warning system (EWS) alert periods. These results demonstrate that average accuracy metrics alone can underestimate the practical value of robust forecasting models.

The findings highlight the potential of foundation models to enhance power system resilience. By leveraging transferable temporal representations learned through large-scale pretraining, Chronos maintains strong predictive performance without task-specific retraining, even under unseen and extreme demand conditions. This robustness is particularly valuable for system operators, for whom forecasting reliability is most critical during periods of elevated operational risk.

Several limitations should be acknowledged. First, the analysis focuses on a single national-level electricity demand dataset; although the zero-shot setting strengthens generalization claims, further validation across diverse power systems is required. Second, this study evaluates forecast accuracy rather than downstream operational or economic impacts, and only a zero-shot deployment scenario is considered.

Future work may extend this study by evaluating foundation models across multiple regions, incorporating probabilistic forecasting and uncertainty quantification, and assessing economic and reliability outcomes. Hybrid approaches that combine pretrained foundation models with limited domain-specific adaptation also represent a promising direction.

Throughout this study, large language models were used in a supportive role to help structure the evaluation framework, surface alternative analytical perspectives, and audit interpretive consistency. All experimental design choices, numerical analyses, and conclusions remain the responsibility of the authors.

By demonstrating that zero-shot models maintain reliability precisely when traditional methods fail, this work establishes foundation models as valuable tools for managing increasingly volatile power systems under climate change.

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309 **A Experimental Setup and Reproducibility**

310 **A.1 Computing Environment**

311 All experiments were conducted in a standardized Python environment. The computing setup is
312 summarized as follows:

- 313 • **CPU:** 13th Gen Intel(R) Core(TM) i7-1360P (2.20 GHz)
- 314 • **GPU:** Not used (Inference conducted on CPU)
- 315 • **RAM:** 16.0 GB
- 316 • **Operating System:** Windows 11 (64-bit)
- 317 • **Python Version:** 3.13.2

318 **A.2 Software and AI Model Specifications**

319 The following core libraries and AI models were used throughout the experiments:

- 320 • **Software Libraries:** numpy (1.26.4), pandas (2.2.2), pmdarima (2.0.4), prophet (1.1.5)
- 321 • **Foundation Model:** Chronos-t5-small (Version: amazon/chronos-t5-small) via Hug-
322 gingFace
- 323 • **Primary LLM:** GPT-4o (Version: gpt-4o-2024-05-13), Temperature: 0.7
- 324 • **Audit/Verification Models:** Gemini 1.5 Pro and Claude 3.5 Sonnet (Multi-panel audit)

325 **A.3 Rolling Forecast Evaluation Protocol**

326 The rolling forecast procedure applies the following steps consistently to all evaluated models:

327 [htbp] 24-hour-ahead Rolling Forecast Evaluation [1] each hour t in the evaluation period Use all
328 available data up to time t as historical context (720-hour window) Generate forecasts for hours $t + 1$
329 to $t + 24$ Record prediction errors for all horizons Aggregate errors over the full evaluation period

330 **A.4 AI-Human Collaborative Workflow**

331 To ensure transparency and reproducibility as required by the 2026 AI Co-Scientist Challenge
332 guidelines, the interaction protocol between the authors and AI agents is detailed below.

333 **A.4.1 Multi-Panel Audit Logic**

334 In accordance with the "Multi-model Audit (3+ models)" requirement, GPT-5.2 was utilized for
335 initial framework proposal (e.g., defining operational regimes), while Gemini 3 Pro and Claude 4.5
336 Sonnet acted as independent scientific auditors to verify the physical plausibility of thresholds and
337 consistency of data interpretations.

338 **A.4.2 Representative Prompts**

339 The following core prompts were used to facilitate the research process:

- 340 • **Framework Construction:** "Define 'operational stress regimes' for the South Korean
341 electricity grid based on 30-day rolling demand changes. Propose thresholds that capture
342 top 5% volatility."
- 343 • **Interpretation Audit:** "Analyze the statistical differences in MAE between Chronos and
344 Prophet during the 2024 heatwave. Provide alternative explanations for performance diver-
345 gence."

346 **A.4.3 Scientific Judgment and Responsibility**

347 While AI models supported documentation and hypothesis refinement, all experimental code imple-
348 mentation, parameter tuning, and final numerical validations were performed strictly by the authors
349 to prevent automated result generation and ensure data integrity. Detailed interaction logs and prompt
350 histories are archived in the separate *AI Usage Report*.

AI Co-Scientist Challenge Korea Paper Checklist

1. Claims

Question: Do the main claims made in the abstract and introduction accurately reflect the paper’s contributions and scope?

Answer: [Yes]

Justification: The abstract and introduction claim (i) a fair and consistent 24-hour-ahead rolling-origin evaluation across all models, and (ii) an empirical finding that a zero-shot time-series foundation model (Chronos) improves short-term electricity demand forecasting accuracy on South Korea’s national hourly load data, with larger gains under operationally critical conditions (operating regimes, high-change events, and early warning system alert periods). These claims directly match the study design and are supported by the quantitative results and figures reported in Section 4.

2. Limitations

Question: Does the paper discuss the limitations of the work performed by the authors?

Answer: [Yes]

Justification: Section 5 and Section 6 explicitly discuss several limitations, including (i) the focus on a single national-level aggregated load dataset, (ii) limited incorporation of exogenous covariates within the forecasting models, (iii) sensitivity of conditional slices (e.g., top 1% high-change events) to threshold definitions, and (iv) the absence of fine-tuning or task-specific adaptation beyond the intended zero-shot setting. While temperature variables are used to define conditional operating regimes (e.g., ExtremeWeather), they are not directly incorporated as model inputs, consistent with the study’s focus on evaluating zero-shot robustness. Concrete future work directions are provided, including multi-region validation, covariate-integrated baselines, probabilistic forecasting, and operational calibration.

3. Theory Assumptions and Proofs

Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

Answer: [N/A]

Justification: This paper is an empirical evaluation study and does not introduce new theoretical results, assumptions, or formal proofs.

4. Experimental Result Reproducibility

Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper?

Answer: [Yes]

Justification: Section 2 specifies the data sources (Korea Power Exchange), temporal coverage (2021–2024 hourly), train–evaluation split (2021–2023 as historical context and 2024 as out-of-sample test), and the exact 24-hour-ahead rolling-origin evaluation protocol. Section 3 details all forecasting models and configurations. Appendix A provides the evaluation procedure in algorithmic form and documents the computing environment, enabling full reproduction using publicly available data and standard software libraries.

5. Open access to data and code

Question: Does the paper provide open access to the data and code?

Answer: [No]

Justification: The study uses publicly available electricity demand data from the Korea Power Exchange (KPX) and meteorological data from the Korea Meteorological Administration (KMA), both accessible through official government portals cited in the paper. The source code is not released with the submission in order to preserve review anonymity, as releasing implementation details during the review phase could compromise anonymization given the combination of national data sources and experimental design. Nevertheless, the paper provides comprehensive methodological details sufficient for reproduction. The code will be made publicly available upon acceptance.

6. Experimental Setting/Details

Question: Does the paper specify all the training and test details necessary to understand the results?

Answer: [Yes]

Justification: All models are evaluated under an identical 24-hour-ahead rolling-origin procedure with a fixed forecast horizon. For ARIMA and Prophet, the update and refitting strategies under an expanding historical window are explicitly described. Chronos is applied strictly in a zero-shot manner without fine-tuning. Evaluation metrics (MAE, RMSE, MAPE) are clearly defined and reported consistently across all experiments.

7. Experiment Statistical Significance

Question: Does the paper report error bars or other information about statistical significance?

Answer: [Yes]

Justification: The paper reports paired statistical tests comparing rolling-origin forecast errors between Chronos and baseline models, demonstrating that observed improvements are statistically significant (paired t -tests with $p < 0.001$). Where appropriate, variability across forecast origins is summarized in tables and figures.

8. Experiments Compute Resources

Question: Does the paper provide sufficient information on the computer resources needed to reproduce the experiments?

Answer: [Yes]

Justification: Appendix A reports detailed compute information, including operating system, CPU usage, absence of GPU acceleration, memory, Python version, and key software dependencies. The experiments are designed to be reproducible on standard personal computing hardware.

9. Code Of Ethics

Question: Does the research conducted in the paper conform to the NeurIPS Code of Ethics?

Answer: [Yes]

Justification: The study uses only aggregated national electricity demand and meteorological data without personal, sensitive, or identifiable information. No human subjects or protected attributes are involved.

10. Broader Impacts

Question: Does the paper discuss potential societal impacts?

Answer: [Yes]

Justification: The paper discusses how improved short-term demand forecasting can enhance power system reliability, operational efficiency, and resilience during extreme events. Potential risks, such as overreliance on automated forecasts, are acknowledged, and the need for human oversight in operational deployment is emphasized.

11. Safeguards

Question: Does the paper describe safeguards for responsible release of data or models?

Answer: [Yes]

Justification: The paper does not introduce or release new sensitive datasets or models. It recommends human-in-the-loop monitoring, conservative alert thresholds, and out-of-distribution awareness for any operational early-warning deployment, and clearly states the non-causal scope of the analysis.

12. Licenses for existing assets

Question: Are the creators or original owners of assets properly credited?

Answer: [Yes]

451 **Justification:** All external datasets, software libraries, and pretrained models (including
452 Chronos, Prophet, and ARIMA implementations) are appropriately cited and credited in the
453 main text and references.

454 13. **New Assets**

455 **Question:** Are new assets introduced and documented?

456 **Answer:** [No]

457 **Justification:** The paper does not introduce new datasets, models, or benchmarks; it presents
458 an empirical evaluation using existing public data and pretrained models.

459 14. **Crowdsourcing and Research with Human Subjects**

460 **Question:** Does the paper include details on crowdsourcing or human-subject research?

461 **Answer:** [N/A]

462 **Justification:** No crowdsourcing and no human-subject research are involved.

463 15. **Institutional Review Board (IRB) Approvals**

464 **Question:** Does the paper describe IRB approvals or equivalent?

465 **Answer:** [N/A]

466 **Justification:** Not applicable, as the study does not involve human participants or identifiable
467 personal data.